# UV divergence of $N_2$ (Group I, Row III) following Balitsky–Chirilli and Lublinsky–Mulian: subtraction in the Wilson lines only, dimensional regularisation of the subtraction term

## 1 Starting expression

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\\
&\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left.\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\\
&\times\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y)+N_c\,U^{af}(x)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y).
\end{aligned}
\tag{1.1}
$$

From here on we use $\xi = p^+/k^+$, so

$$
\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,\frac{1}{p^+(y-x)^2+(k^+-p^+)(y-z)^2}=\int_{\Lambda/k^+}^{1-\Lambda/k^+}\frac{d\xi}{2\pi}\,\frac{1}{\xi(y-x)^2+(1-\xi)(y-z)^2}.
$$

## 2 Step 1: subtraction in the Wilson lines only

(Balitsky–Chirilli eq. (59); Lublinsky–Mulian eqs. (4.71)–(4.73).) The only UV divergence is at $z\to x$, where the kernel goes as $1/(x-z)^2$. Following Balitsky–Chirilli and Lublinsky–Mulian, we subtract and add the configuration in which the two gluons cross the shockwave at the *same* point. Only the Wilson lines are changed; the kernel, *including the exponential*, is untouched. Setting $z\to x$ in the colour factor and using

$$
f^{abc}\,f^{def}\,U^{be}(x)\,U^{cf}(x)=N_c\,U^{ad}(x)
\tag{2.1}
$$

gives $-f^{adc}f^{fbe}U^{db}(x)U^{ce}(x)+N_cU^{af}(y)=N_c\big[U^{af}(y)-U^{af}(x)\big]$, the single-gluon colour structure. Hence

$$
\frac{d^3N_2}{d^3k}=\underbrace{\left(\frac{d^3N_2}{d^3k}-\frac{d^3N_2^{\mathrm{sub}}}{d^3k}\right)}_{[-ffU(x)U(z)+N_cU(x)]}+\underbrace{\frac{d^3N_2^{\mathrm{sub}}}{d^3k}}_{N_c[U(y)-U(x)]}\quad,
\tag{2.2}
$$

with the same kernel in both terms. This is exactly the split $N_2=N_{2a}+N_{2b}$.

- $N_2-N_2^{\mathrm{sub}}$ is UV finite, because its colour factor vanishes at $z=x$. It is computed in $d=2$.
- $N_2^{\mathrm{sub}}$ carries the UV divergence. Its colour factor does not depend on $z$, so the $z$ integral of the *full kernel* is done in $d_\perp=2-\epsilon$.

# 3 Step 2: exact split of the subtraction kernel

Contract with $(x'-w')^i$. The bracket of (1.1), multiplied by $\frac{(z-x)^m}{(z-x)^2}$, is identically equal to (UV part) + (rest):

$$\text{UV:}\quad d_\perp\,\xi(1-\xi)\,\frac{(z-x)^i\,(z-x)\cdot(y-x)}{(z-x)^4}+\Big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\frac{(y-x)^i}{(z-x)^2}, \tag{3.1}$$

$$\begin{aligned}\text{rest:}\quad &\frac{(z-x)^i}{(z-x)^2}\Big(-\frac{d_\perp}{2}\xi(1-\xi)+2(1-\xi)+\frac{\xi}{2}-(1-\xi)\frac{(y-x)\cdot(z-x)}{2(y-x)^2}-\xi\frac{(y-x)\cdot(y-z)}{2(y-z)^2}\Big)\\&+\frac{\xi}{1-\xi}\frac{(y-x)^i\,(z-x)\cdot(y-z)}{2(z-x)^2(y-z)^2}-\frac{1-\xi}{\xi}\Big(\frac{(z-x)^i}{(z-x)^2}+\frac{(z-x)\cdot(y-x)\,(y-z)^i}{2(z-x)^2(y-x)^2}\Big)\\&-\frac{(z-x)^i}{(z-x)^2}+\frac{(z-x)\cdot(y-x)\,(y-z)^i}{2(z-x)^2(y-z)^2}-\frac{(y-x)^i\,(z-x)\cdot(y-z)}{2(z-x)^2(y-x)^2}.\end{aligned} \tag{3.2}$$

This was checked numerically at random points for $d_\perp = 2.0,\ 2.3,\ 2.6,\ 2.9$; the residual is $10^{-15}$. The rest is $O(1/|z-x|)$, so it is UV finite and taken in $d=2$. The UV part contains the whole UV divergence. Its angle-averaged coefficient is $\frac{d_\perp}{d}\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}$, which equals $\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}$ for $d_\perp = d$. We keep $d_\perp = d$ in the UV part.

# 4 Step 3: dimensional regularisation of the UV part; the exponential becomes a momentum shift

**3a. Energy denominator in momentum space.** By the four-dimensional identity $\int d^4P\, e^{iP\cdot R}/P^2 = 4\pi^2/R^2$, after rescaling $p\to\sqrt{\xi}\,p$, $l\to\sqrt{1-\xi}\,l$:

$$\frac{1}{\xi(y-x)^2+(1-\xi)(y-z)^2}=\frac{1}{4\pi^2\,\xi(1-\xi)}\int d^2p\,d^2l\,\frac{e^{\,ip\cdot(y-x)+il\cdot(y-z)}}{\frac{p^2}{\xi}+\frac{l^2}{1-\xi}}. \tag{4.1}$$

**3b. The exponential.** Since $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\xi}=e^{-ik\cdot(w'-x)}\,e^{i(1-\xi)k\cdot(z-x)}$ and $e^{il\cdot(y-z)}=e^{il\cdot(y-x)}e^{-il\cdot(z-x)}$, the whole $z$ dependence of the UV part is

$$\int d^dz\,e^{-i\,(l-(1-\xi)k)\cdot(z-x)}\Big[d_\perp\xi(1-\xi)\frac{(z-x)^i(z-x)^j}{(z-x)^4}(y-x)^j+\Big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\frac{(y-x)^i}{(z-x)^2}\Big]. \tag{4.2}$$

The exponential is therefore not reduced or approximated. It just shifts the momentum at which the kernel is Fourier transformed, to $l-(1-\xi)k$. In Balitsky–Chirilli's dipole case there is no external momentum, so there is no shift.

**3c. The $d$-dimensional Fourier transforms** ($d=2-\epsilon$, $\overline{MS}$, $\bar\mu^2=4\pi e^{-\gamma_E}\mu^2$):

$$\mu^{-\epsilon}\int d^dr\,\frac{e^{-iQ\cdot r}}{r^2}=\pi^{d/2}\,2^{d-2}\,\Gamma\big(\tfrac{d}{2}-1\big)\,(Q^2)^{1-\frac{d}{2}}\mu^{-\epsilon}=\pi\Big[\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{Q^2}\Big], \tag{4.3}$$

$$\mu^{-\epsilon}\int d^dr\,e^{-iQ\cdot r}\,\frac{r^ir^j}{r^4}=\frac{\delta^{ij}}{d}\,\pi\Big[\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{Q^2}\Big]-\pi\Big(\frac{Q^iQ^j}{Q^2}-\frac{\delta^{ij}}{2}\Big). \tag{4.4}$$

The traceless part follows from $\int d\varphi\,e^{-iQr\cos(\varphi-\varphi_Q)}\cos2\varphi=-2\pi J_2(Qr)\cos2\varphi_Q$ and $\int_0^\infty J_2(x)\,dx/x=\frac12$. As a check, with a short-distance cut-off $\delta$, $\int d^2r\,e^{-iQ\cdot r}/r^2=2\pi[-\log(|Q|\delta/2)-\gamma_E]$, and the replacement $\log\frac{1}{\delta^2}\to\frac{2}{\epsilon}+2\gamma_E+\log\frac{\bar\mu^2}{4}$ reproduces the first line.

**3d. Result for the UV part** ($d_\perp = d$):

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{UV}}^{\mathrm{sub}}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda/k^+}^{1-\Lambda/k^+}\frac{d\xi}{2\pi}\int_{x,y,x',w'}e^{-ik\cdot(w'-x)}\\&\times\frac{1}{4\pi^2\,\xi(1-\xi)}\int d^2p\,d^2l\,\frac{e^{\,i(p+l)\cdot(y-x)}}{\frac{p^2}{\xi}+\frac{l^2}{1-\xi}}\,\frac{1}{(x'-w')^2}\\&\times\Bigg\{\pi\Big(\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)(x'-w')\cdot(y-x)\left[\frac{2}{\epsilon}+\log\frac{\bar\mu^2}{\big(l-(1-\xi)k\big)^2}\right]\\&\quad-2\pi\,\xi(1-\xi)\left[\frac{(x'-w')\cdot\big(l-(1-\xi)k\big)\,\big(l-(1-\xi)k\big)\cdot(y-x)}{\big(l-(1-\xi)k\big)^2}-\frac{(x'-w')\cdot(y-x)}{2}\right]\Bigg\}\\&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\,N_c\Big[U^{af}(y)-U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y).\end{aligned} \tag{4.5}$$

If desired, the $p$ integral can be done: $\int d^2p\,\frac{e^{ip\cdot(y-x)}}{p^2/\xi+l^2/(1-\xi)}=2\pi\xi\,K_0\big(|l|\,|y-x|\sqrt{\xi/(1-\xi)}\big)$.

The equivalent coordinate-space form (with the $l$ integral done using a Feynman parameter and Bessel functions $K_0, K_1, K_2$) is in `N2b_UV_z_integrated_explicit.pdf`. The two forms were checked against each other numerically; they agree to the accuracy of the momentum grid ($2 \times 10^{-3}$).

**3e. The UV pole and the** $\log \bar{\mu}$**.** Undoing the $p, l$ integral at fixed scale gives $1/(y-x)^2$. So the $\frac{2}{\epsilon}$ and the $\log \bar{\mu}^2$ multiply

$$\pi\Big(\xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}\Big)\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}.$$

With $\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi \left(\xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}\right) = 2\log\frac{k^+-\Lambda}{\Lambda} - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}$, the coefficient of $\log \bar{\mu}^2$ is

$$\frac{1}{(2\pi)^3}\frac{g^4 N_c}{16\pi^4}\frac{1}{k^+}\Big[2\log\frac{k^+}{\Lambda} - \frac{11}{6}\Big]\int_{x,y,x',w'} e^{-ik\cdot(w'-x)}\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(x)\Big]\rho^{b'}(x')\rho^f(y). \tag{4.6}$$

In the convention $\int d^2p/p^2 = 0$ the $\frac{2}{\epsilon}$ is dropped. The $-\frac{11}{6}\log\bar{\mu}^2$ is the running-coupling term, which combines with Group I Row II. The $2\log\frac{k^+}{\Lambda}\log\bar{\mu}^2$ goes with the JIMWLK terms.

# 5 Step 4: the finite pieces

**4a.** $N_2 - N_2^{\rm sub}$ ($d=2$, $d_\perp = 2$): the full expression (1.1) with colour factor

$$\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_c U^{af}(x)\Big]\rho^{b'}(x')\rho^f(y).$$

**4b. Rest of the subtraction term** ($d=2$, $d_\perp = 2$):

$$\frac{d^3N_{2\,\rm rest}^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{\Lambda/k^+}^{1-\Lambda/k^+}\frac{d\xi}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\xi}$$
$$\times\frac{1}{\xi(y-x)^2 + (1-\xi)(y-z)^2}\,\frac{(x'-w')^i}{(x'-w')^2}$$
$$\times\Bigg[\frac{(z-x)^i}{(z-x)^2}\Big(-\xi(1-\xi) + 2(1-\xi) + \frac{\xi}{2} - (1-\xi)\frac{(y-x)\cdot(z-x)}{2(y-x)^2} - \xi\frac{(y-x)\cdot(y-z)}{2(y-z)^2}\Big)$$
$$+\frac{\xi}{1-\xi}\frac{(y-x)^i\,(z-x)\cdot(y-z)}{2(z-x)^2(y-z)^2} - \frac{1-\xi}{\xi}\Big(\frac{(z-x)^i}{(z-x)^2} + \frac{(z-x)\cdot(y-x)\,(y-z)^i}{2(z-x)^2(y-x)^2}\Big)$$
$$-\frac{(z-x)^i}{(z-x)^2} + \frac{(z-x)\cdot(y-x)\,(y-z)^i}{2(z-x)^2(y-z)^2} - \frac{(y-x)^i\,(z-x)\cdot(y-z)}{2(z-x)^2(y-x)^2}\Bigg]$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\,N_c\Big[U^{af}(y) - U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{5.1}$$

# 6 Summary

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = \underbrace{\Big(\frac{d^3N_2}{d^3k} - \frac{d^3N_2^{\rm sub}}{d^3k}\Big)}_{\text{finite, } d=2} + \underbrace{\frac{d^3N_{2\,\rm UV}^{\rm sub}}{d^3k}}_{\text{eq. (4.5), dim. reg.}} + \underbrace{\frac{d^3N_{2\,\rm rest}^{\rm sub}}{d^3k}}_{\text{finite, } d=2}. \tag{6.1}$$

- As in BC and LM, only the Wilson lines are changed by the subtraction; the exponential stays in the kernel. In the $d$-dimensional $z$ integral it becomes the momentum shift $l \to l - (1-\xi)k$. No extra subtraction in $z$ is made.
- The only $\log\bar{\mu}$ is in eq. (4.5), with coefficient $\frac{g^4N_c}{16\pi^4}\big[2\log\frac{k^+}{\Lambda} - \frac{11}{6}\big]$.
- **Next step: the plus prescription in** $\xi$ on the finite pieces and on the remaining $\xi$ dependence of eq. (4.5).
- **Caution 1.** In the rest, the term $-\frac{1-\xi}{\xi}\frac{(z-x)^i}{(z-x)^2}$ gives a spurious divergence at $z \to y$ in the $\xi \to 0$ endpoint, where the denominator becomes $(y-z)^2$. It cancels against the UV part. To avoid it, move this term into the UV part, where it turns $\frac{1-\xi}{\xi}\frac{(y-x)^i}{(z-x)^2}$ into $\frac{1-\xi}{\xi}\frac{(y-z)^i}{(z-x)^2}$. The UV properties of both parts are unchanged.
- **Caution 2.** Near $\xi \to 1$ the momentum shift vanishes, $(1-\xi)k \to 0$, and the $\xi \to 1$ endpoint has large-distance (infrared) behaviour. This shows up as $\frac{1}{2}\log^2\frac{k^+}{\Lambda}$ in the coordinate-space form. It must cancel against the other $\log(k^+/\Lambda)$ (JIMWLK) terms; it is not a UV effect.